

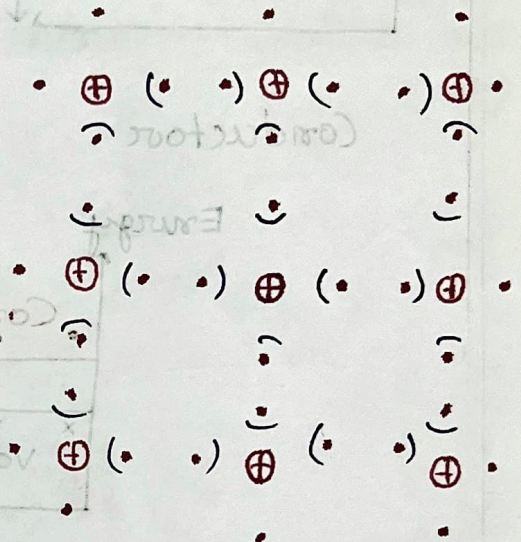
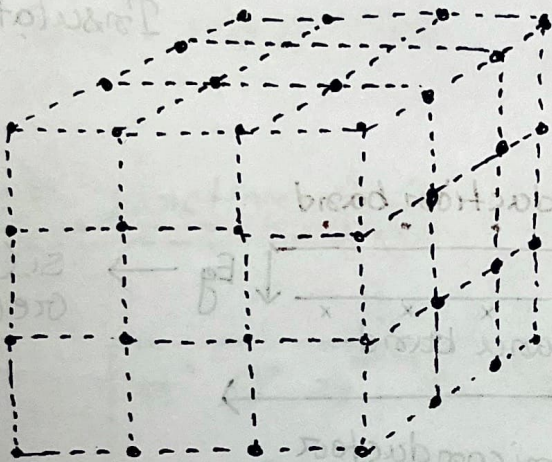
Semiconductor

- જાતિ (Material) - ① નીચીદારી (Conductor)
 ② અર્ધનીચીદારી (Semiconductor)
 ③ ઇસુલેટર (Insulator)

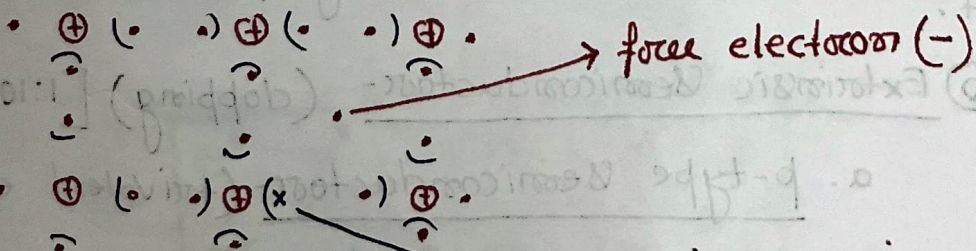
→ Conductor → free electron.

→ Semiconductor → Si, Ge.

→ Lattice Structure -



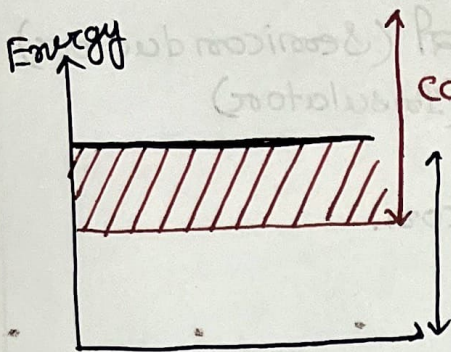
there is no free electron or charge carrier in any semiconductor at normal temperature.



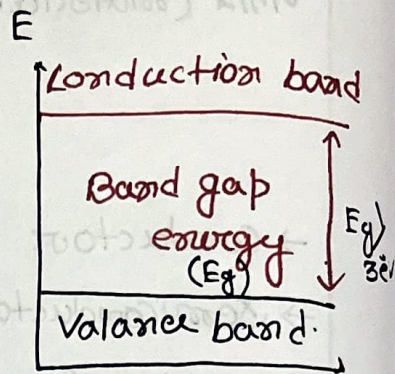
there are two types of charge carrier in semiconductor
 ① free electron and ② hole.

* free electron can move through any place but hole can only move through the covalent bond.

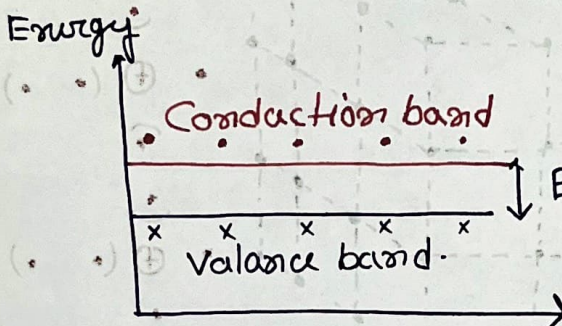
* Band diagram-



Conductor



Insulator



Semiconductor

Si (1.1 eV)
Ge (0.67 eV)

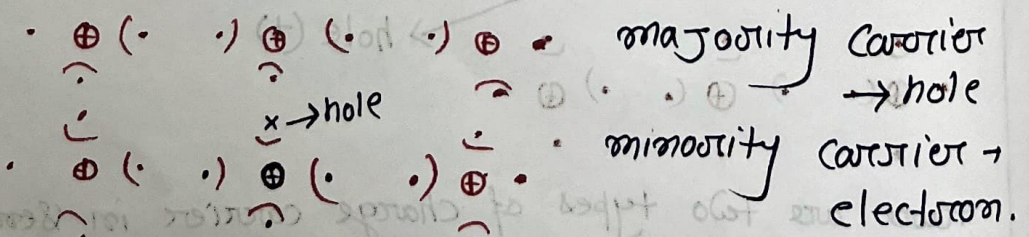
* Semiconductor types -

① Intrinsic Semiconductor.

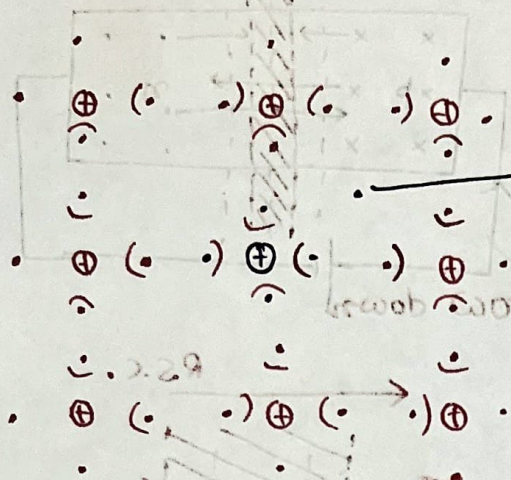
the number of generated electron-hole pairs is equal. There is only pure semiconductor exist.

② Extrinsic Semiconductor - (doping) [1:1000, 1:100]

a. p-type Semiconductor - (trivalent element)



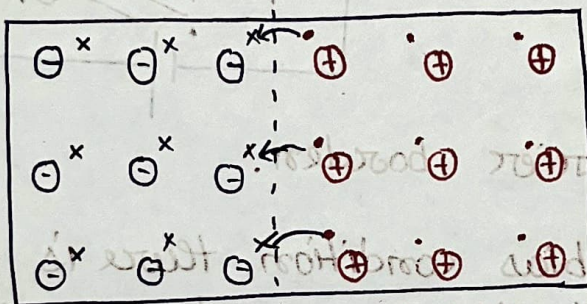
b. n type Semiconductor - (penta Valent element)



free electron = majority carrier.
(nA)

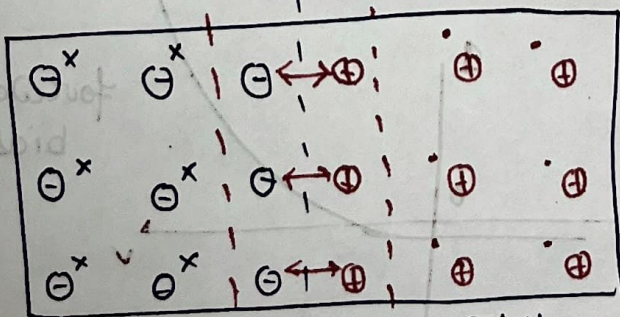
minority carrier = hole.
(pA)

* p-n Junction diode -



p type n type

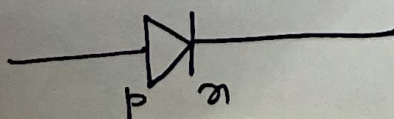
the combination of electron and hole $\rightarrow$ recombination.



p type

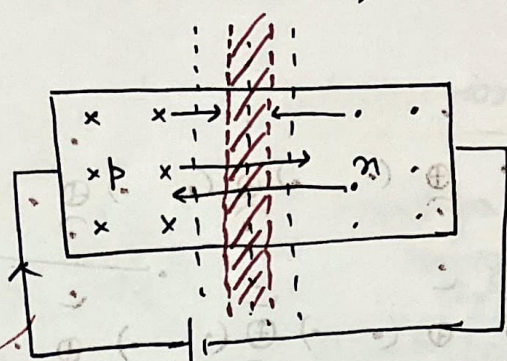
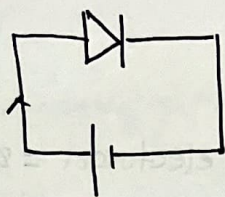
n type

Potential barrier.



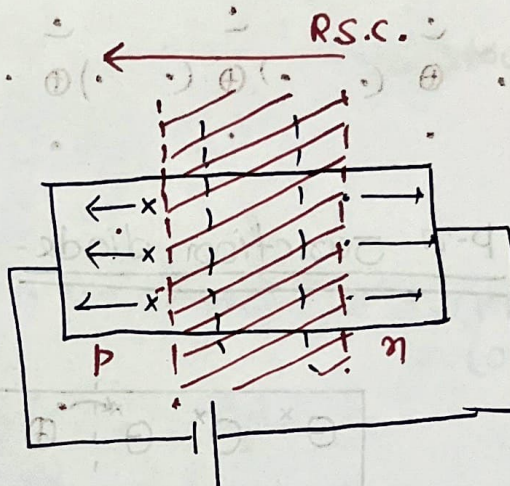
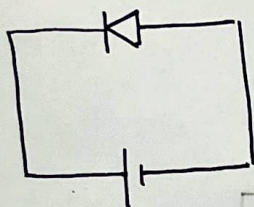
* Forward biasing -

current flow.



potential barrier narrow down

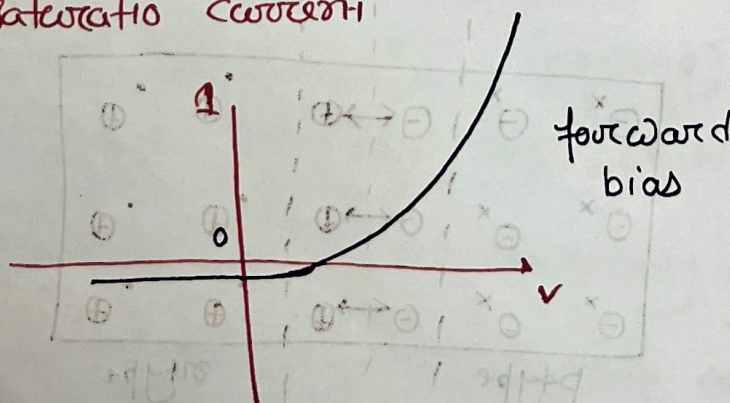
* Reverse biasing -



potential barrier broaden.

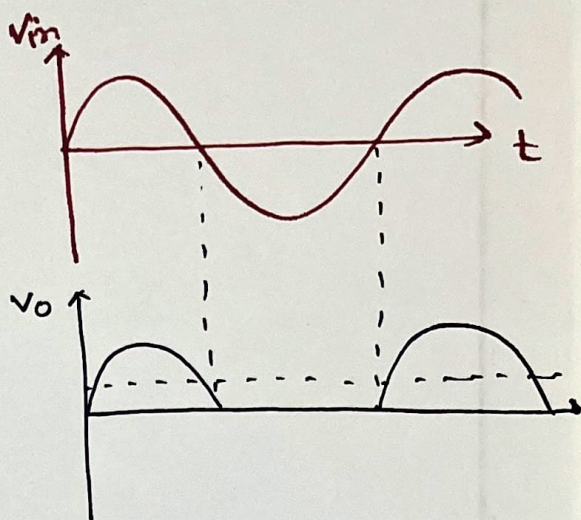
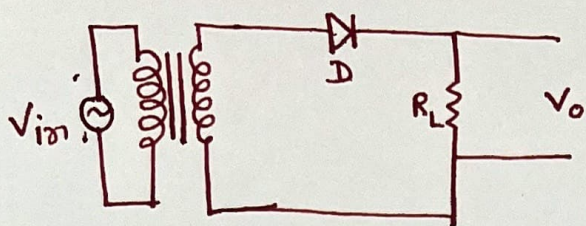
In reverse bias condition there is no current flow due to the majority carrier.

But there is current flow in (uA) order due to minority carrier. This is known as reverse saturation current.



Reverse bias.

* Half wave rectifier-



* Full wave rectifier-

